BIOS 731: Final Report

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Spring 2026

Abstract

Traditional transcriptome-wide and proteome-wide association studies (TWAS/PWAS) limit analyses to *cis*- molecular quantitative trait loci (xQTL), neglecting *trans*-xQTL that compose a substantial proportion of molecular variation. The BGW-xWAS tool implements a Bayesian-Genome Wide (BGW) framework based upon a Bayesian variable selection regression (BVSR) model that jointly accounts for both *cis*- and *trans*-xQTL factors to improve genetically-regulated gene expression (GReX) and protein abundance (GReP) imputation accuracy and xWAS power over *cis*-only methods. To study both *cis*- and *trans*- regulatory mechanisms of AD, this study applies the BGW-xWAS tool to train xWAS models of dorsolateral prefrontal cortex tissue from ROS/MAP and integrates the results with large genome-wide association study (GWAS) summary data of Alzheimer disease (AD) dementia in order to identify genetically-regulated molecular associations underlying AD risk.

1 Introduction

Traditional xWAS analyses neglect the importance of distant xQTL factors (i.e., *trans*-eQTLs located outside of the ±1 MB region) that possess a key role in molecular variation. In highly polygenic neurodegenerative disease (e.g., Alzheimer's disease (AD)), genome-wide TWAS analyses that jointly model *cis*- and *trans*-eQTLs have shown that roughly one-quarter to one-third of significant AD risk genes across brain and blood are predominantly driven by *trans*-eQTL weights. This includes 34 of 141 genome-wide significant genes in a recent Bayesian TWAS of AD dementia, indicating that *cis*-only models miss a substantial fraction of disease-relevant regulatory effects [1].

The BGW-TWAS tool [2] was developed in order to circumvent the computational burden of incorporating *trans*- variants into TWAS analyses. In this paper the underlying models used by the BGW-xWAS tool are described. An application TWAS and PWAS of AD is conducted to study the impacts of the *cis*- and *trans*- regulatory mechanisms of AD dementia using the BGW-xWAS tool to account for *cis*- and *trans*-xQTL factors to improve genetically regulated gene expression (GReX) imputation accuracy and xWAS power over *cis*-only methods [2].

1.1 Standard Two-Stage xWAS

The standard two-stage xWAS methods [3, 4, 5, 6] first fit gene quantitative trait (genetically-regulate gene expression (GReX), genetically-regulated protein abundance (GReP)) prediction models using the reference -omic and genetic data profiled for the same samples, and then test the association between the predicted genetically regulated quantitative trait (GReX or GReP) and phenotype of interest for the test GWAS cohort. Given a reference genotype, $\mathbf{G}_0$, and quantitative trait (gene expression, protein abundance, etc.), $\mathbf{Q}_0$, data from the same subjects,

$$\mathbf{Q}_0 = \mathbf{G}_0\mathbf{w} + \boldsymbol{\epsilon}, \qquad \boldsymbol{\epsilon} \sim N(0,\, \sigma_\epsilon^2\mathbf{I})$$

where $\mathbf{w}$ is xQTL weights. In step 2, the association between predicted genetically-regulated quantitative trait (GReX or GReP) and phenotype of interest xQTL weights $\hat{\mathbf{w}}$ estimated in step 1:

$$\widehat{\mathbf{GReX}} = \mathbf{G}_{\text{test}}\hat{\mathbf{w}}$$

2 BGW-xWAS Tool

Although *trans*-SNPs explain much variation in expression quantitative traits, and may be significant themselves, existing TWAS tools only considered *cis*-SNPs (variants within ±1MB of a gene). Including *trans*- variants expected to increase imputation accuracy of GReX and the power of TWASs, however, routine TWAS methods are impractical given the computational cost required to fit 20K GReX imputation models for genomewide genes and 10M genotypes per tissue. The BGW-TWAS tool [2] was developed in order to circumvent this computational burden. While it was originally developed for TWAS, the same software can be applied to other kinds of QTLs; in this context, the tool is referred to as "BGW-xWAS".

In order to reduce computational burden, BGW-xWAS employs a scalable EM-MCMC algorithm and uses summary statistics of standard eQTL analyses based on single variant tests. Dependencies include C++ libraries zlib, gsl, eigen3, lapack, atlas, blas, and libStatGen[7]. For computational efficiency, it is recommended these libraries and scripts for the tool itself be compiled by the user.

2.1 Bayesian variable selection regression (BVSR)

The eQTL effect size w_i follows a "spike-and-slab" prior distribution–ie, we only retain a subset of the possible regressors; w_i follows a $N(0, \sigma_w^2\sigma_\epsilon^2)$ distribution with probability π and w_i follows point-mass density function $\partial_0(\cdot)$ (ie, is 0) with probability $1-\pi$. If $w_i = 0$ then $\partial_0(w_i) = 1$, otherwise $\partial_0(w_i) = 0$.

$$\mathbf{E}_{g,n\times 1} = \mathbf{X}_{n\times p}\mathbf{w}_{p\times 1} + \boldsymbol{\epsilon}_{n\times 1}, \qquad w_i \sim \pi N(0, \sigma_w^2\sigma_\epsilon^2) + (1-\pi)\partial_0(\cdot), \qquad \epsilon_i \sim N(0, \sigma_\epsilon^2)$$

The BVSR model is expanded to include both *cis*- and *trans*- variants:

$$\mathbf{E}_g = \mathbf{X}_{\text{cis}}\mathbf{w}_{\text{cis}} + \mathbf{X}_{\text{trans}}\mathbf{w}_{\text{trans}} + \boldsymbol{\epsilon}, \qquad \epsilon_i \sim N(0, \sigma_\epsilon^2)$$
$$w_{\text{cis},i} \sim \pi_{\text{cis}} N(0, \sigma_{\text{cis}}^2\sigma_\epsilon^2) + (1-\pi_{\text{cis}})\partial_0(w_{\text{cis},i}), \qquad w_{\text{trans},i} \sim \pi_{\text{trans}} N(0, \sigma_{\text{trans}}^2\sigma_\epsilon^2) + (1-\pi_{\text{trans}})\partial_0(w_{\text{trans},i})$$

The model assumes independent conjugate hyper priors:

$$\pi_{\text{cis}} \sim Beta(a_{\text{cis}}, b_{\text{cis}}), \qquad \sigma_{\text{cis}}^2 \sim IG(k_1, k_2) \qquad \sigma_\epsilon^2 \sim IG(k_3, k_4)$$
$$\pi_{\text{trans}} \sim Beta(a_{\text{trans}}, b_{\text{trans}}), \qquad \sigma_{\text{trans}}^2 \sim IG(k_3, k_4)$$

To further reduce computational burden, a latent indicator vector $\boldsymbol{\gamma}_{p\times 1}$ is introduced to aide computation. Each element γ_i indicates whether $w_{\cdot,i} = 0$ ($\gamma_i = 0$) or follows the $N(0, \sigma_\cdot^2\sigma_\epsilon^2)$ distribution ($\gamma_i = 1$). $\text{E}(\gamma_i)$ is the posterior probability (PP_i) for the ith SNP to be an eQTL with non-zero effect size w_i.

2.2 Efficient Computation Using EM-MCMC

While eQTL effect sizes and posterior probabilities $(\hat{w}_i, \widehat{PP}_i)$ could be obtained with a standard Markov Chain Monte Carlo (MCMC) algorithm, this is not feasible for genome-wide modeling due to slow convergence rate and large memory usage. Instead, BGW-TWAS further adapts an existing scalable expectation maximization EM-MCMC to use only summary statistics, incliding linkage disequillibrium and score statistics from standard eQTL analyses by single variant tests. Genotype regions are further pruned to reduce computational burden.

During the E-step, the EM-MCMC algorithm uses MCMC per genome block (in parallel) to estimate $(\mathbf{w}, \mathbf{PP})$ from most recent estimation of $(\boldsymbol{\pi}, \boldsymbol{\sigma}^2)$. During the M-step, the newly obtained $(\boldsymbol{\pi}, \boldsymbol{\sigma}^2)$ is used to update $(\boldsymbol{\pi}, \boldsymbol{\sigma}^2)$. Iterations continue until maximum a posteriori (MAP) estimates converge. Once the final estimates $(\hat{\mathbf{w}}, \widehat{\mathbf{PP}})$ are obtained, estimated GReX is calculated:

$$\widehat{\mathbf{GReX}} = \sum_{i=1}^{p} \tilde{X}_i(\widehat{PP}_i\hat{w}_i)$$

2.3 Summary-Level Association Test

A burden Z-score as implemented by S-PrediXcan[8] can be used to test the association between predicted GReX and a phenotype of interest.

$$\widetilde{Z}_g = \sum_{i \in \text{Model}} \frac{(\widehat{PP_i}\,\widehat{w_i})\,\widehat{\sigma_i}\,Z_{\text{GWAS},i}}{\sqrt{(\widehat{\vec{PP}}\,\widehat{\vec{w}})'\,\vec{V}\,(\widehat{\vec{PP}}\,\widehat{\vec{w}})}}, \qquad \widehat{\sigma_i^2} = \text{Var}(\vec{G}_{0,i}), \qquad \vec{V} = \text{Cov}(\vec{G}_0)$$

where $\widehat{PP_i}$ is the estimated posterior causal probability for SNP i, $\widehat{w_i}$ is theestimated xQTL effect-size for SNP i,$Z_{\text{GWAS},i}$ is the Z-score statistic values for single variant test (ie, GWAS), $\widehat{\vec{PP}}$ is the estimated posterior causal probabilities from trained model,$\widehat{\vec{w}}$ is the xQTL effect-sizes from trained model, $\vec{V}$ is the genotype covariance matrix (LD data) estimated from reference panel genotype data $\vec{G}_0$, $\vec{G}_0$ is the reference genotype matrix, and $\widehat{\sigma_i}$ is the estimated standard deviation of SNP i from genotype vector for SNP i, $\vec{G}_{0,i}$.

3 Methods

3.1 ROS/MAP Dataset

The Religious Orders Study (ROS) and Rush Memory and Aging Project (MAP) [9] are two similarly-structured longitudinal prospective clinical-pathologic cohort studies of aging and AD. These studies are collectively referred to as ROS/MAP. RNA-seq (N_{subjects}=931, N_{genes}=21,244) and tandem mass tag (TMT) proteomics data (N_{subjects}=716, N_{genes}=12,175) of postmortem dorsolateral prefrontal cortex (DLPFC) tissue of ROS/MAP were used in xWAS training.

3.2 AD GWAS Dataset

After training, the estimated xQTL weights were integrated with AD GWAS data from Bellenguez et al. (2022). This GWAS study had a sample size of 111,326 clinically diagnosed/"proxy" AD cases and 677,663 controls.

4 Results

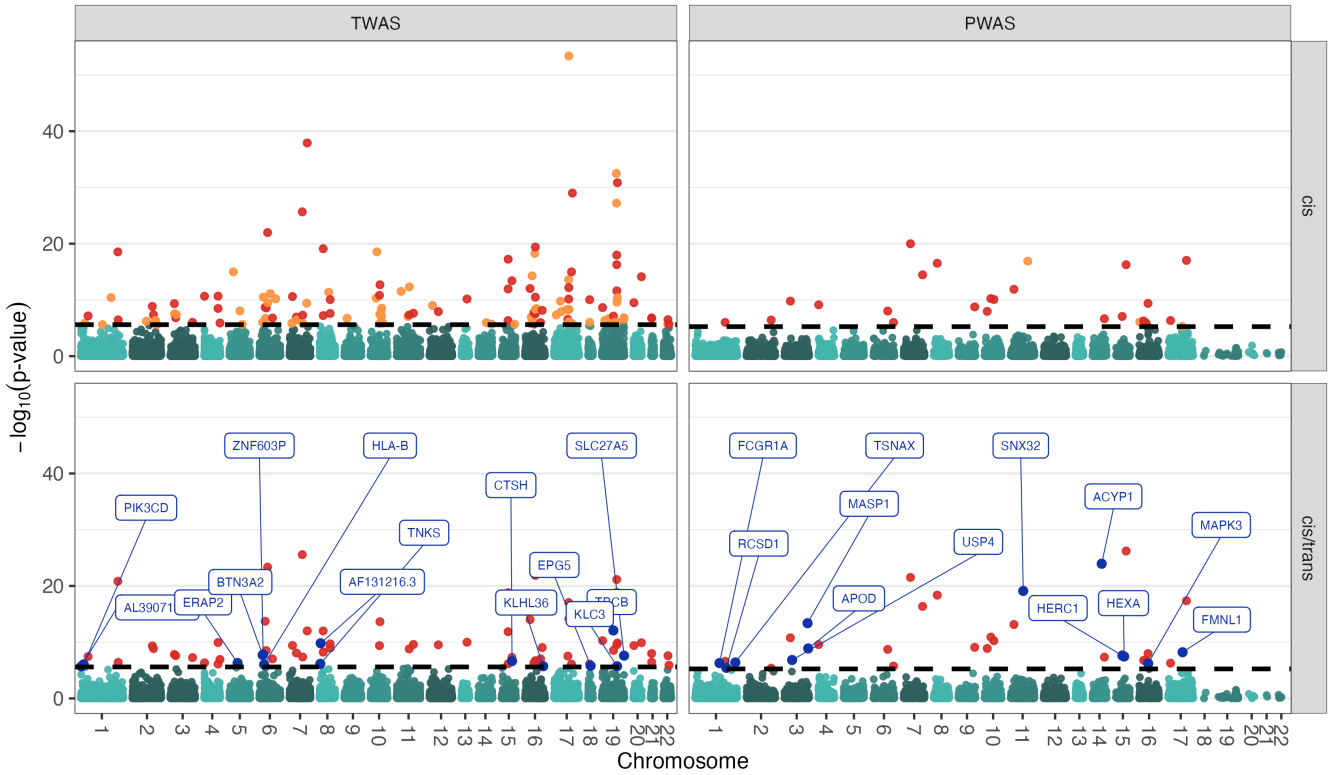
For the TWAS models, we obtained GReX models for 16,408 and 16,676 genes for *cis*- and *cis*/*trans*-TWAS, respectively. For the PWAS models, we obtained GReP models for 8,829 and 8,922 genes for *cis*- and *cis*/*trans*-TWAS, respectively. These results are shown in Table 1. Additionally, the *cis*/*trans*-xWAS methods obtained some *trans*-only gene models: 298 for TWAS and 84 for PWAS.

Table 1: xWAS Association Test Results Summary

	Total Genes	Significant	Nonsignificant	NA	*trans*-only
cis-TWAS	16408	130	15973	305	0
cis/trans-TWAS	16676	74	16047	555	298
cis-PWAS	8829	25	7553	1251	0
cis/trans-PWAS	8922	34	7595	1293	84

Manhattan plots of the xWAS study results shown in Figure 1. The xWAS studies identified a total of 263 xWAS significant risk genes of AD: 130 *cis*-TWAS, 74 *cis*/*trans*-TWAS, 25 *cis*-PWAS, and 34 *cis*/*trans*-PWAS risk genes. These significant risk genes are shown in Tables 5 to 8. Table 4 shows genes identified only by the *cis*/*trans*-xWASs.

Figure 1: Manhattan Plots of xWAS Results



Significant genes identified by only the *cis*/*trans* xWASs are labeled and shown in blue. Significant genes identified by both the *cis*- and *cis*/*trans* models for a given xWAS (TWAS or PWAS) are shown in red. All other significant genes are shown in orange. Non-significant genes are shown in green.

Figure 2 shows a venn diagram of overlapping risk genes identified by these four xWAS studies: 4 genes (*ACE*, *EGFR*, *EPHX2*, *LACTB*) were identified by all four studies and another 5 (*CTSH*, *EARS2*, *FCER1G*, *LARS2*, *SNX32*) were identified by at least three of the four xWASs. With the exception of known AD TWAS risk gene *LARS2* [11], these genes have been previously identified as GWAS risk genes of AD. These risk, as well as a brief description of any known functions, is in Table 2. Other previously identified GWAS and TWAS risk genes of AD that have been replicated here include *CR1*, *HLA-DRA*, and *MAPT-AS1* [12].

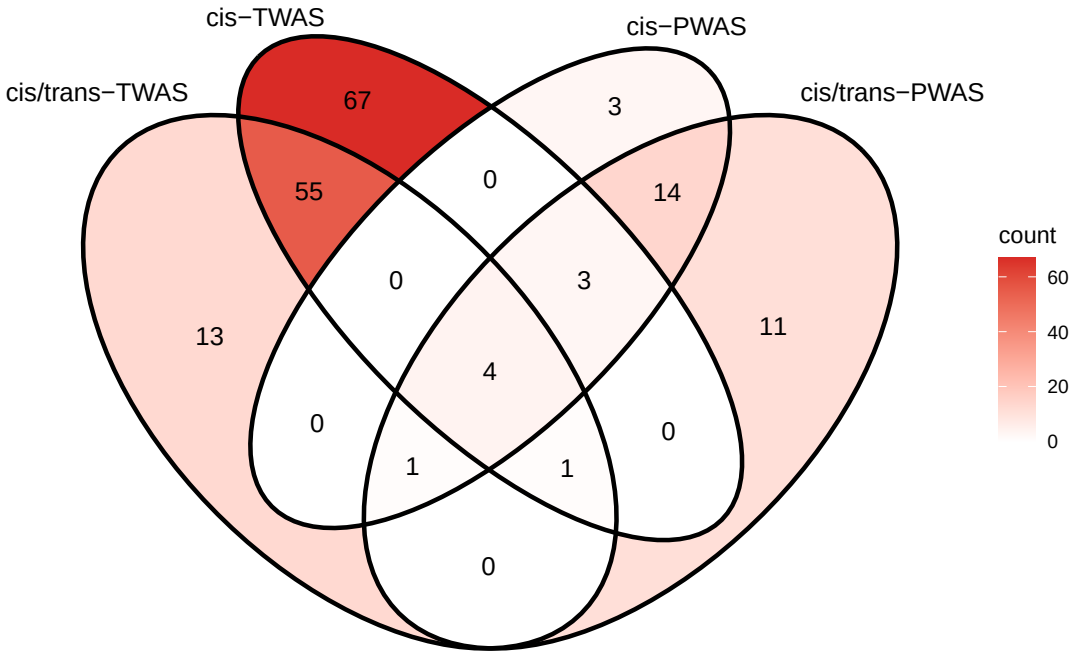
Table 2: Risk Genes Identified by 3 or more xWAS

CHR	Gene	Function/Disease Association
1	*FCER1G**	immunoreceptor
3	*LARS2***	catalyzes attachment of amino acid Leu to tRNA
7	*EGFR**	epidermal growth factor receptor
8	*EPHX2**	lipid mediator; mutations associated with familial hypercholesterolemia
11	*SNX32**	may be involved intercellular trafficking
15	*LACTB**	regulator of mitochondrial lipid metabolism
15	*CTSH**	degradation of proteins in lysosomes
16	*EARS2**	catalyzes attachment of amino acids Glu and Gln to mitochondrial tRNAs
17	*ACE**	angiotensin-converting enzyme

∗ previously-identified GWAS risk genes

∗∗ TWAS risk gene of AD, potential drug target [11]

Figure 2: Venn Diagrams of Overlapping xWAS Results



PathDIP [13] was used to perform enrichment analyses on the risk genes obtained from the four xWAS. Significantly enriched pathways for each are shown in Figure 3 while Figure 4 shows the number of significant enriched pathways shared across the xWASs; there were 5 pathways significantly enriched for all xWASs (Table 3).

5 Summary

Traditional xWAS studies neglect *trans*-QTLs which can explain a significant amount of variation of expression quantitative traits and explain underlying biological mechanisms, but the associated computational burden renders routine methods non-feasible. The BGW-xWAS tool was developed to reduce computational burden of incorporating *trans*-QTLs into the standard two-stage xWAS model. Application studies of AD xWAS show risk genes significantly enriched for AD-related pathways, including Alzheimer disease-amyloid secretase path.

BGW-TWAS does have some limitations: it assumes a two-stage model for xWASs and computational costs are still substantial to train models for genome-wide genes per tissue type, per -omic data type: it takes approximately 30min of computation time per gene (in parallel with 4 CPU cores). BGW-PWAS considers only protein abundance and does not consider the effect of genetic variants on protein function [14].

6 References

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Appendices

A Figures

Figure 3: Significant Enriched Paths by pathDIP

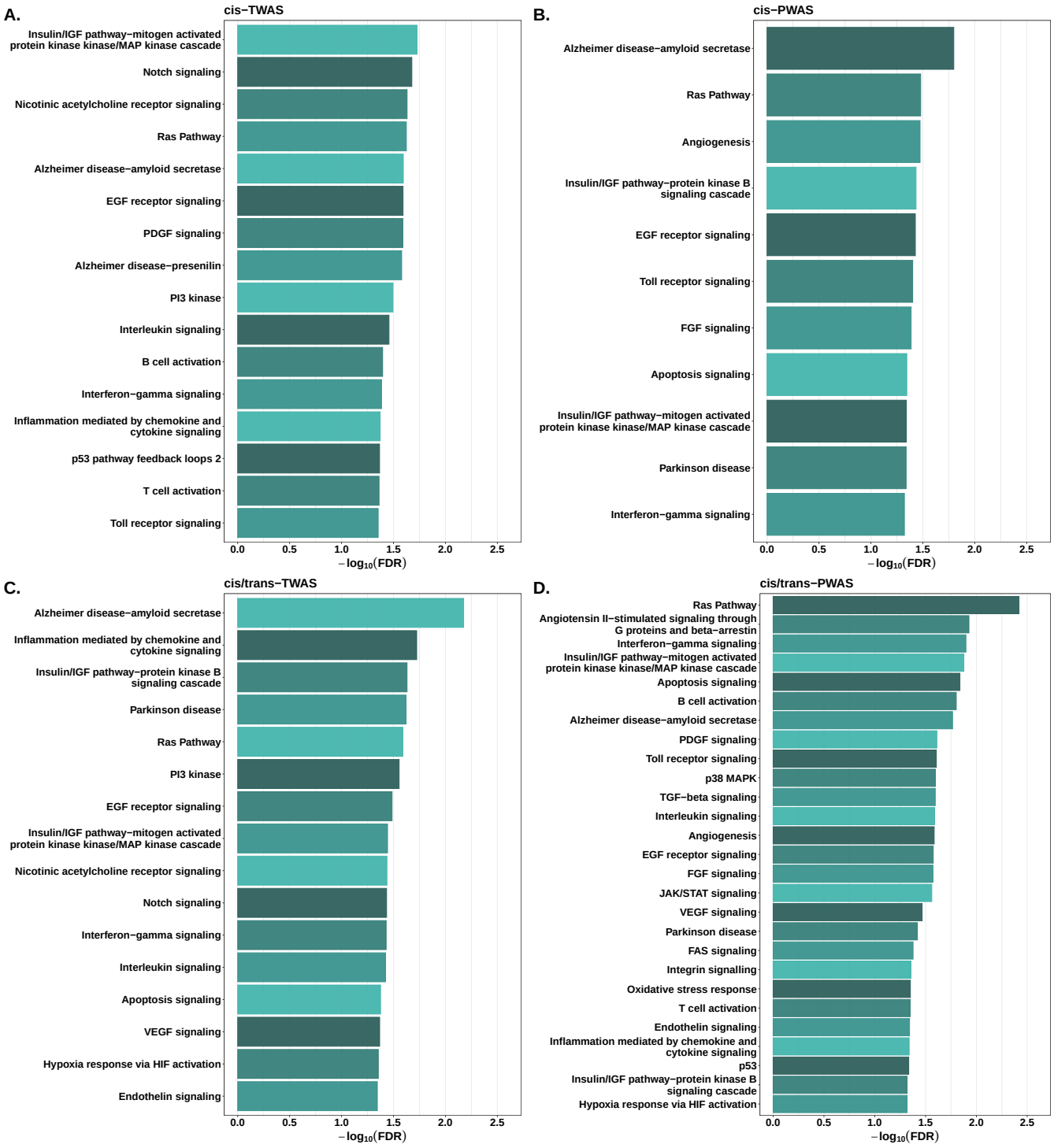
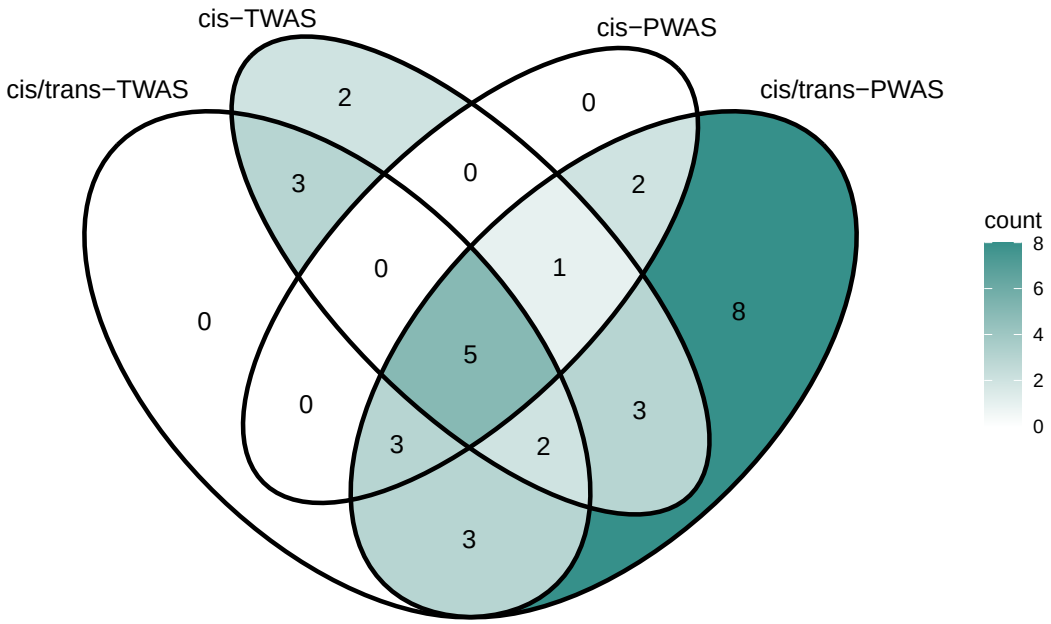


Figure 4: Venn Diagram of Number of Overlapping Paths for each xWAS



B Tables

Table 3: Significantly Enriched Pathways for all xWAS

Alzheimer disease-amyloid secretase
EGF receptor signaling
Insulin/IGF pathway-mitogen activated protein kinase kinase/MAP kinase cascade
Interferon-gamma signaling
Ras Pathway

Table 4: Risk Genes Identified Only by *cis*/*trans*-xWAS

CHR	GeneName	TWAS	PWAS	Function
1	*FCGR1A*	0	1	IgG immune response
1	*PIK3CD**	1	0	immune response mediation
1	*RCSD1*	0	1	actin filament binding activity
1	*TSNAX*	0	1	involved in RNA-induced silencing complex
3	*APOD**	0	1	lipid transport
3	*MASP1*	0	1	complement system
3	*USP4*	0	1	ubiquitinating enzyme
5	*ERAP2*	1	0	peptide trimming for HLA class 1-binding peptides
6	*BTN3A2**	1	0	T-cell responses
6	*HLA-B*	1	0	major histocompatibility complex class I
8	*AF131216.3**	1	0	lncRNA
8	*TNKS*	1	0	involved in telomere length, vesicle trafficking
11	*SNX32**	1	1	may be involved in intracellular trafficking
14	*ACYP1***	0	1	-
15	*CTSH*	1	1	degradation of proteins in lysosomes
15	*HERC1***	0	1	involved in membrane trafficking
16	*KLHL36**	1	0	may be involved in ubiquitination
16	*MAPK3*	0	1	MAP kinase signal transduction pathway
17	*FMNL1***	0	1	regulation of cell morphology
18	*EPG5*	1	0	involved in autophagy
19	*KLC3**	1	0	play a role in organelle transport
19	*SLC27A5*	1	0	fatty acid formation
19	*TBCB***	1	0	tubulin folding

$*$ previously-identified GWAS risk gene of AD

$**$ previously-identified GWAS risk gene of dementia

Table 5: AD risk genes identified by cis-TWAS

CHR	Start	End	Gene	p-value
1	207496147	207640647	*CR1*	2.85e-19
1	161215234	161220699	*FCER1G*	3.73e-11
1	26169871	26170873	*ZNF593*	6.79e-08
1	209938174	210171389	*SYT14*	3.49e-07
1	12019466	12032045	*MIIP*	1.33e-06
1	111473792	111478512	*C1orf162*	2.44e-06
2	174431571	174487094	*GPR155*	1.40e-09
2	186590065	186680901	*ITGAV*	4.32e-08
2	202773150	202871985	*ICA1L*	4.64e-07
2	200973718	201071671	*FAM126B*	5.41e-07
2	108534355	108687246	*LIMS1*	5.86e-07
2	218255319	218257366	*AC021016.2*	2.35e-06
3	36826820	36945057	*TRANK1*	4.57e-10
3	37243177	37366751	*GOLGA4*	2.71e-08
3	45388506	45549421	*LARS2*	3.94e-08
3	42979267	43060211	*FAM198A*	1.48e-07
3	184230429	184242329	*VWA5B2*	9.59e-07
4	158170752	158202877	*AC098679.1*	2.14e-11
4	784957	826198	*CPLX1*	2.21e-11
4	158124474	158173318	*FAM198B*	3.28e-09
4	185399540	185425985	*UFSP2*	1.26e-06
5	56909260	56925532	*SETD9*	1.02e-15
5	131245493	131398447	*CDC42SE2*	8.67e-09
5	140332843	140346631	*HBEGF*	2.18e-06
6	41158506	41163186	*TREM2*	1.03e-22
6	61630233	61681049	*AL049544.1*	7.45e-12
6	29555629	29633976	*GABBR1*	3.14e-11
6	125747664	125761269	*HEY2*	6.30e-11
6	42184401	42194916	*GUCA1B*	3.13e-10
6	32439842	32445046	*HLA-DRA*	2.07e-09
6	34587288	34696859	*C6orf106*	2.64e-09
6	89568144	89638753	*LYRM2*	1.64e-07
6	32934629	32941070	*HLA-DMB*	1.96e-07
6	26224199	26227473	*HIST1H3E*	9.48e-07
7	135662496	135693418	*C7orf73*	1.22e-38
7	100367530	100400099	*PILRA*	2.15e-26
7	21543215	21901839	*DNAH11*	2.50e-11
7	130930209	130932206	*AC016831.5*	3.87e-10
7	100679507	100689448	*GIGYF1*	5.13e-08
7	55019021	55211628	*EGFR*	1.12e-07
7	55951793	56000181	*NIPSNAP2*	3.66e-07
7	8113184	8262687	*ICA1*	1.41e-06
7	100826820	100867009	*SLC12A9*	1.60e-06
8	27490779	27545564	*EPHX2*	7.85e-20
8	89901859	89927888	*OSGIN2*	4.34e-12
8	100572882	100663415	*SNX31*	8.60e-11
8	100685816	100722809	*PABPC1*	2.53e-08

8	27633868	27676776	*SCARA3*	6.04e-08
9	34638133	34651035	*GALT*	1.80e-07
10	68414064	68472121	*DNA2*	2.74e-19
10	87094161	87191468	*FAM35A*	1.98e-13
10	80454166	80533123	*TSPAN14*	1.52e-11
10	58513140	58831437	*BICC1*	4.83e-11
10	97332497	97334709	*FRAT2*	2.92e-09
10	95663396	95694143	*TCTN3*	6.34e-09
10	86668449	86736068	*LDB3*	3.77e-08
10	97356358	97426076	*RRP12*	7.46e-08
10	96191702	96271587	*BLNK*	7.50e-07
10	100969458	100987515	*MRPL43*	1.34e-06
10	86654557	86666848	*OPN4*	1.50e-06
11	65883741	65888539	*FIBP*	4.92e-13
11	18322295	18367044	*GTF2H1*	3.00e-12
11	78435961	78574874	*NARS2*	2.33e-08
11	65833641	65856896	*SNX32*	4.70e-08
11	64340223	64357534	*CCDC88B*	8.55e-08
12	9277235	9313241	*AC009533.1*	9.84e-10
12	51351247	51392867	*GALNT6*	1.19e-08
13	94601903	94634645	*GPR180*	6.79e-11
14	70325081	70359731	*COX16*	1.08e-06
14	103694560	103695170	*AL049840.3*	1.98e-06
15	63276018	63309126	*APH1B*	5.69e-18
15	78565520	78595269	*CHRNA5*	3.85e-14
15	63121800	63142061	*LACTB*	1.10e-12
15	63042632	63071915	*TPM1*	4.78e-07
15	90883695	90895776	*FES*	2.10e-06
16	31138925	31150094	*PRSS36*	3.96e-20
16	30757423	30762710	*CCDC189*	5.26e-19
16	23522014	23557731	*EARS2*	5.21e-15
16	11927373	11976643	*NPIPB2*	9.45e-13
16	31214021	31231537	*TRIM72*	3.20e-11
16	53433977	53491649	*RBL2*	3.40e-09
16	31215962	31217359	*PYDC1*	4.64e-09
16	77247813	77435114	*ADAMTS18*	7.13e-09
16	29995294	30005793	*INO80E*	1.80e-08
16	31115754	31131393	*KAT8*	3.16e-08
16	19701934	19718235	*KNOP1*	1.07e-07
16	30075978	30085377	*PPP4C*	9.56e-07
16	70246778	70251930	*EXOSC6*	1.08e-06
16	19761172	19766099	*AC027130.1*	1.65e-06
17	46516702	46579682	*ARL17A*	4.18e-54
17	63477061	63498380	*ACE*	9.98e-30
17	58345175	58353093	*SUPT4H1*	1.03e-15
17	45506741	45550335	*LRRC37A4P*	2.42e-14
17	46511511	46555650	*LRRC37A2*	6.53e-13
17	45799390	45895630	*MAPT-AS1*	6.99e-11
17	18108706	18179802	*MYO15A*	1.70e-10
17	46984045	47100323	*AC068152.1*	4.86e-09

17	42659305	42667006	*TUBG2*	5.08e-09
17	45051610	45109016	*NMT1*	5.86e-09
17	20008051	20009234	*AC005730.2*	1.15e-08
17	4967992	4987652	*CAMTA2*	4.66e-08
17	44345086	44353102	*GRN*	2.80e-07
17	46978481	47054569	*LRRC37A17P*	7.45e-07
17	58325450	58415766	*TSPOAP1-AS1*	1.37e-06
17	45620344	45835826	*LINC02210-CRHR1*	1.52e-06
18	23506184	23586898	*NPC1*	9.09e-11
18	24138956	24161603	*CABYR*	9.46e-07
19	43872363	43901385	*ZNF404*	3.20e-33
19	45782947	45792802	*DMWD*	1.43e-31
19	44051367	44067991	*ZNF223*	6.22e-28
19	44662278	44684359	*CEACAM19*	1.06e-18
19	44643798	44663583	*PVR*	5.33e-17
19	44809059	44821420	*BCAM*	2.42e-12
19	45714387	45717381	*AC007191.1*	3.97e-11
19	45079288	45305283	*MARK4*	2.46e-10
19	1086579	1095380	*POLR2E*	2.27e-09
19	36313067	36331718	*LINC00665*	7.64e-08
19	58404238	58408484	*AC012313.1*	1.42e-07
19	15640897	15662825	*CYP4F3*	2.09e-07
19	58432814	58440222	*ZNF132*	2.69e-07
19	5784832	5791238	*DUS3L*	4.29e-07
19	44846175	44889228	*NECTIN2*	9.05e-07
19	36334453	36379199	*ZFP14*	1.06e-06
20	63861212	63864293	*AL118506.1*	7.44e-15
20	1563521	1620061	*SIRPB1*	3.07e-10
21	26170871	26217381	*AP000229.1*	1.56e-07
21	26393635	26569252	*CYYR1-AS1*	1.79e-07
22	37830855	37849327	*ANKRD54*	3.19e-07
22	37180167	37199385	*C1QTNF6*	1.91e-06
22	41091786	41180079	*EP300*	2.16e-06

Table 6: AD risk genes identified by cis/trans-TWAS

CHR	Start	End	Gene	p-value
1	207496147	207640647	*CR1*	1.45e-21
1	26169871	26170873	*ZNF593*	3.58e-08
1	209938174	210171389	*SYT14*	3.91e-07
1	9651732	9729114	*PIK3CD*	8.61e-07
1	1059734	1069355	*AL390719.1*	2.03e-06
2	174431571	174487094	*GPR155*	4.57e-10
2	186590065	186680901	*ITGAV*	1.39e-09
3	36826820	36945057	*TRANK1*	1.53e-08
3	42979267	43060211	*FAM198A*	2.48e-08
3	184230429	184242329	*VWA5B2*	5.44e-08
4	158170752	158202877	*AC098679.1*	1.15e-10
4	185399540	185425985	*UFSP2*	1.17e-07
4	784957	826198	*CPLX1*	4.62e-07
4	158124474	158173318	*FAM198B*	7.73e-07
5	96875939	96919716	*ERAP2*	4.99e-07
6	41158506	41163186	*TREM2*	4.45e-24
6	32439842	32445046	*HLA-DRA*	1.87e-14
6	34587288	34696859	*C6orf106*	3.05e-09
6	28176188	28176674	*ZNF603P*	1.53e-08
6	26365159	26378320	*BTN3A2*	1.93e-08
6	89568144	89638753	*LYRM2*	9.44e-08
6	31269491	31357188	*HLA-B*	9.13e-07
7	100367530	100400099	*PILRA*	2.70e-26
7	135662496	135693418	*C7orf73*	9.81e-13
7	21543215	21901839	*DNAH11*	3.77e-10
7	55019021	55211628	*EGFR*	8.43e-09
7	100679507	100689448	*GIGYF1*	4.30e-08
8	27490779	27545564	*EPHX2*	1.01e-12
8	11345748	11347502	*AF131216.3*	1.57e-10
8	100685816	100722809	*PABPC1*	2.01e-10
8	100572882	100663415	*SNX31*	1.08e-09
8	27633868	27676776	*SCARA3*	5.89e-09
8	9555914	9782346	*TNKS*	7.16e-07
10	87094161	87191468	*FAM35A*	2.30e-14
10	80454166	80533123	*TSPAN14*	4.08e-10
11	78435961	78574874	*NARS2*	2.58e-10
11	65833641	65856896	*SNX32*	1.58e-09
12	51351247	51392867	*GALNT6*	2.96e-10
13	94601903	94634645	*GPR180*	1.00e-10
15	63276018	63309126	*APH1B*	1.55e-19
15	63121800	63142061	*LACTB*	1.36e-12
15	78565520	78595269	*CHRNA5*	5.02e-08
15	78921058	78949574	*CTSH*	2.25e-07
15	63042632	63071915	*TPM1*	7.02e-07
16	31138925	31150094	*PRSS36*	1.37e-22
16	11927373	11976643	*NPIPB2*	9.68e-15
16	77247813	77435114	*ADAMTS18*	8.76e-10

16	70246778	70251930	*EXOSC6*	9.28e-08
16	31214021	31231537	*TRIM72*	1.93e-07
16	31115754	31131393	*KAT8*	2.60e-07
16	29995294	30005793	*INO80E*	4.79e-07
16	84648525	84667686	*KLHL36*	1.89e-06
17	63477061	63498380	*ACE*	9.08e-26
17	45799390	45895630	*MAPT-AS1*	8.69e-18
17	46511511	46555650	*LRRC37A2*	7.09e-15
17	44345086	44353102	*GRN*	2.94e-08
17	58345175	58353093	*SUPT4H1*	8.09e-07
18	23506184	23586898	*NPC1*	7.61e-07
18	45847609	45967274	*EPG5*	1.57e-06
19	44662278	44684359	*CEACAM19*	7.17e-22
19	44643798	44663583	*PVR*	1.34e-19
19	36114289	36125947	*TBCB*	8.20e-13
19	1086579	1095380	*POLR2E*	5.36e-11
19	45782947	45792802	*DMWD*	1.39e-10
19	44809059	44821420	*BCAM*	2.79e-10
19	36313067	36331718	*LINC00665*	2.80e-09
19	58479512	58512413	*SLC27A5*	2.49e-08
19	45333434	45351520	*KLC3*	1.90e-06
20	63861212	63864293	*AL118506.1*	1.25e-10
20	1563521	1620061	*SIRPB1*	3.93e-10
21	26170871	26217381	*AP000229.1*	1.03e-08
21	26393635	26569252	*CYYR1-AS1*	3.51e-07
22	37830855	37849327	*ANKRD54*	2.51e-08
22	41091786	41180079	*EP300*	1.24e-06

Table 7: AD risk genes identified by cis-PWAS

CHR	Start	End	Gene	p-value
1	161215297	161218943	*FCER1G*	9.84e-07
2	218274194	218292557	*TMBIM1*	3.78e-07
3	45388583	45547530	*LARS2*	1.62e-10
4	986997	1004393	*IDUA*	7.41e-10
6	107867756	107958009	*SEC63*	9.52e-09
6	160531483	160666375	*LPA*	1.02e-06
7	55019032	55205617	*EGFR*	1.02e-20
7	149003062	149028408	*PDIA4*	3.41e-15
8	27491002	27544522	*EPHX2*	2.99e-17
9	137077501	137082729	*UAP1L1*	1.74e-09
10	68282660	68332927	*PBLD*	5.75e-11
10	86958531	86962985	*SNCG*	8.28e-11
10	22534837	22714326	*PIP4K2A*	1.08e-08
11	85628573	85636229	*TMEM126B*	1.26e-17
11	10511678	10541227	*RNF141*	1.27e-12
14	96363452	96385684	*GSKIP*	2.19e-07
15	78921750	78944981	*CTSH*	5.50e-17
15	63121800	63141805	*LACTB*	8.79e-08
16	30005514	30013596	*DOC2A*	4.06e-10
16	8797813	8847825	*PMM2*	5.38e-07
16	569968	583634	*PIGQ*	8.09e-07
16	23522013	23557343	*EARS2*	1.23e-06
17	63477061	63497366	*ACE*	9.44e-18
17	5282263	5383223	*RABEP1*	4.62e-07
17	44905626	44915486	*GFAP*	5.29e-06

Table 8: AD risk genes identified by cis/trans-PWAS

CHR	Start	End	Gene	p-value
1	161215297	161218943	*FCER1G*	2.47e-07
1	231528653	231564905	*TSNAX*	3.94e-07
1	149782694	149791517	*FCGR1A*	5.20e-07
1	167630237	167704696	*RCSD1*	3.61e-06
2	218274194	218292557	*TMBIM1*	4.12e-06
3	187216085	187291632	*MASP1*	4.15e-14
3	45388583	45547530	*LARS2*	1.65e-11
3	195568702	195584205	*APOD*	1.29e-09
3	49277144	49340024	*USP4*	1.47e-07
4	986997	1004393	*IDUA*	2.71e-10
6	107867756	107958009	*SEC63*	1.88e-09
6	160531483	160666375	*LPA*	1.75e-06
7	55019032	55205617	*EGFR*	3.05e-22
7	149003062	149028408	*PDIA4*	4.38e-17
8	27491002	27544522	*EPHX2*	4.25e-19
9	137077501	137082729	*UAP1L1*	8.26e-10
10	68282660	68332927	*PBLD*	1.22e-11
10	86958531	86962985	*SNCG*	5.48e-11
10	22534837	22714326	*PIP4K2A*	1.31e-09
11	65833939	65853335	*SNX32*	7.72e-20
11	10511678	10541227	*RNF141*	7.15e-14
14	75053225	75069291	*ACYP1*	1.12e-24
14	96363452	96385684	*GSKIP*	4.61e-08
15	78921750	78944981	*CTSH*	6.11e-27
15	63121800	63141805	*LACTB*	1.77e-08
15	63608618	63833948	*HERC1*	2.47e-08
15	72343437	72375972	*HEXA*	3.55e-08
16	30005514	30013596	*DOC2A*	1.06e-08
16	8797813	8847825	*PMM2*	1.68e-07
16	23522013	23557343	*EARS2*	1.70e-07
16	30114105	30123209	*MAPK3*	6.02e-07
17	63477061	63497366	*ACE*	4.27e-18
17	45221925	45247316	*FMNL1*	6.10e-09
17	5282263	5383223	*RABEP1*	5.30e-07